

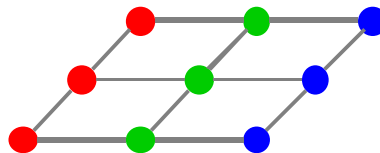
Ontology in Knowledge Representation

William L. Benzon

Abstract. The categorization of entities into classes such as object, plant, animal, or human reflects ontological structure. Ontological structure can be represented by inheritance trees which are orthogonal to more conventional "isa" inheritance trees. Given ontological structure we can define paradigmatic transitions, such as that from caterpillar to butterfly, and ontological transitions, such as that from living to dead. These concepts are exemplified with examples from everyday knowledge and from the world of computer integrated manufacturing. A final section discusses the implications of ontological representation for representation of scientific concepts.

CONTENTS

Introduction: Salt and Sodium Chloride	2
Ontology and Assignment	2
Paradigmatic and Ontological Transition	6
CIMWorld	8
Multiple Ontologies.....	15
References	18



708 Jersey Avenue, 2A
Jersey City, NJ 07302
201.217.1010
bbenzon@mindspring.com

Copyright © 1985, 1987 by William L. Benzon. All Rights Reserved.

Ontology in Knowledge Representation

William L. Benzon

Introduction: Salt and Sodium Chloride

We all know that salt and sodium chloride are, physically, pretty much the same. Conceptually they are very different. Salt is a white granular substance with a certain taste, which is more salient in our understanding of salt than its appearance or texture. After all, the taste tells us of salt's presence even where there is no white granular substance to be seen or touched. Salt is thus rather adequately defined in terms of sensory perceptions.

Sodium chloride is a chemical compound whose molecules consist of one atom of sodium and one atom of chlorine. What in that definition is a sensory perception? "Compound," "atom," "molecule," these are all abstract. And if we start looking behind these abstractions one route will lead us to meter readings on laboratory instruments while another route will lead us to hadrons, leptons, and a handfull of forces, strong, weak, electromagnetic, and gravitational. The conceptual domain in which we find "sodium chloride" is thus quite different from the one in which we find "salt." To borrow from the language of set theory, the extensions of those terms are much the same (note that salt ordinarily contains impurities which aren't in the extension of "sodium chloride") but their intensions are different.

"Salt" is defined in the domain of every-day commonsense knoweldge while "sodium chloride" is defined in the specialized domain of chemistry. These domains have different ontologies. This paper is about the representation of ontology, and I stress "representation of." Philosophers and physicists do worry a great deal about ontological questions - what is the world ultimately made of? Except to the extent that the mind/body problem is properly the concern of artificial intelligence or cognitive science, questions of ontology are not central to us. But, the representation of knowledge is central, and ontological knowledge is clearly important in thought.

Frank Keil's *Semantic and Conceptual Development* [22] provides strong experimental evidence for the existence of ontological categorization as a distinct component of human thought. The model I present in the following discussion is consistent with his model in broad outline, but differs in details. After a general discussion of ontological structure I move to a set of examples chosen from CIMWorld, a knowledge representation being developed at RPI in connection with the Computer Integrated Manufacturing (CIM) Project. I then conclude by moving to broader issues, considering some of the work on naive models of physical phenomena and the role of ontology in scientific thought.

Ontology and Assignment

Aristotle propounded a view in which, to state it so compactly that much subtlety is lost, objects consist of form and substance, plants of an object and a vegetative soul, animals of a plant and a sensitive soul, and human beings of an animal and a rational soul. To this I would add that a social person consists of a human being and a social role (Figure 1). The resulting tree is an assignment (ASN) tree.

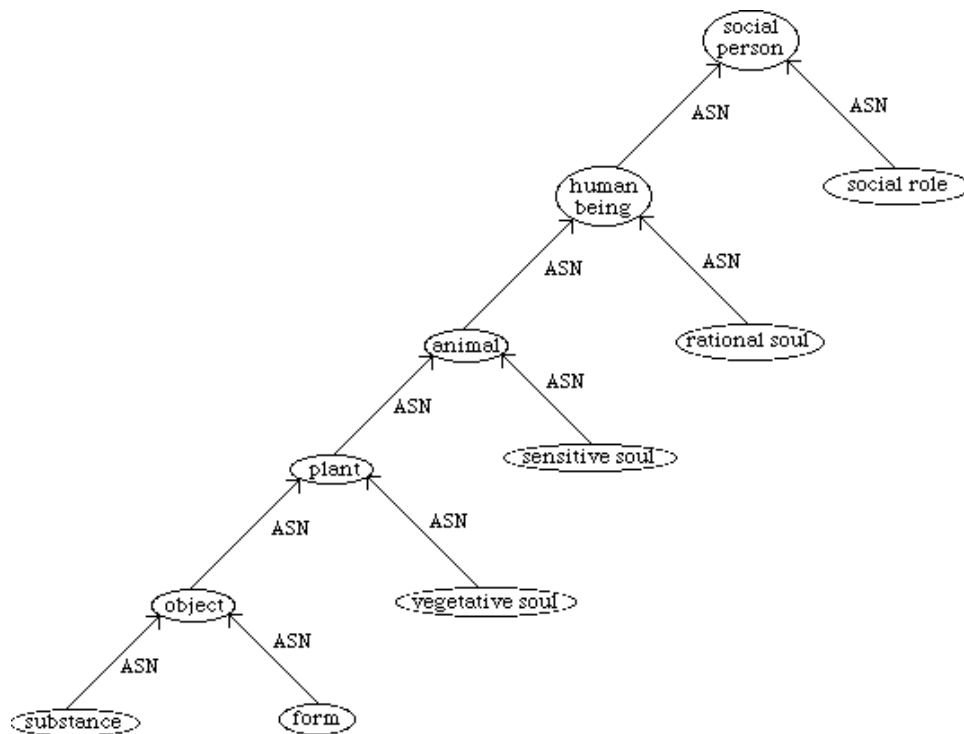


Figure 1: Assignment and Ontology

Each node in the tree designates a different ontological domain, with domains being inherited up assignment arcs. Thus anything which can sensibly be predicated of objects in a subordinate domain can also be predicated of objects in the superordinate domain, but not vice versa. Since shapes can be rounded or jagged, so can objects, plants, animals, and humans, but not souls. Since substances can be heavy or light, hot or cold, so can objects, plants, and the rest. Since animals can be hungry or sleepy, so can humans. But it makes no sense to talk of hungry objects or sleepy shapes. And talk of hungry or sleepy plants, while more acceptable, is obviously metaphorical. Hunger and sleep are activities appropriate to sensitive, but not vegetative, souls.

The nature of ontological structure can be clarified by contrasting it with paradigmatic structure. Figure 2 gives a paradigmatic structure of animals, a typical inheritance hierarchy in which subordinates inherit the characteristics of superordinates, unless explicitly contradicted.

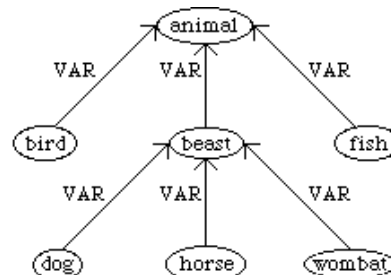


Figure 2: Paradigmatic Tree

Thus dog, horse, and wombat are varieties (VAR) of beast and bird, beast, and fish are varieties of animal. Just as dog, horse, and wombat inherit the characteristics of beast - head, tail, trunk, four legs, fur, etc. - so bird, beast, and fish inherit the characteristics of animal, of which its

position in the assignment structure may be the most important. But why do we need two different inheritance structures?

As Winograd [36] pointed out, one problem which inheritance systems have is inheritance from multiple superordinates. The problem becomes particularly acute when we consider social structure. With only paradigmatic structure available, student, teacher, mother, son, waitress, judge, etc. all become varieties of person and any instance of person is likely to be an instance of several different varieties of social person (Figure 3).

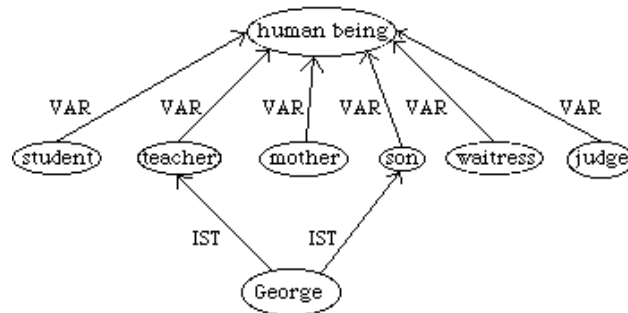


Figure 3: Social Person

Instance (IST) is a paradigmatic arc and thus instance nodes (at the tail end of the arc) can inherit properties and structure from their varietal superordinates (compare the notion of Individual Concept in [7]).

Consider George, who is both a teacher and a son. Teachers are supposed to wield authority while sons are supposed to submit to it. As an exemplary teacher and son George is thus both submissive and authoritative, which doesn't quite make sense. We know that George is not, in fact, both, but that he only acts according to the demands of the role he is currently playing. But paradigmatic inheritance doesn't give us a natural means of expressing this, differentiating between what inheres in George and what inheres in the role he plays. By treating the relationship between George and his various roles as one of assignment this problem is eliminated, because George inherits nothing from the roles he plays.

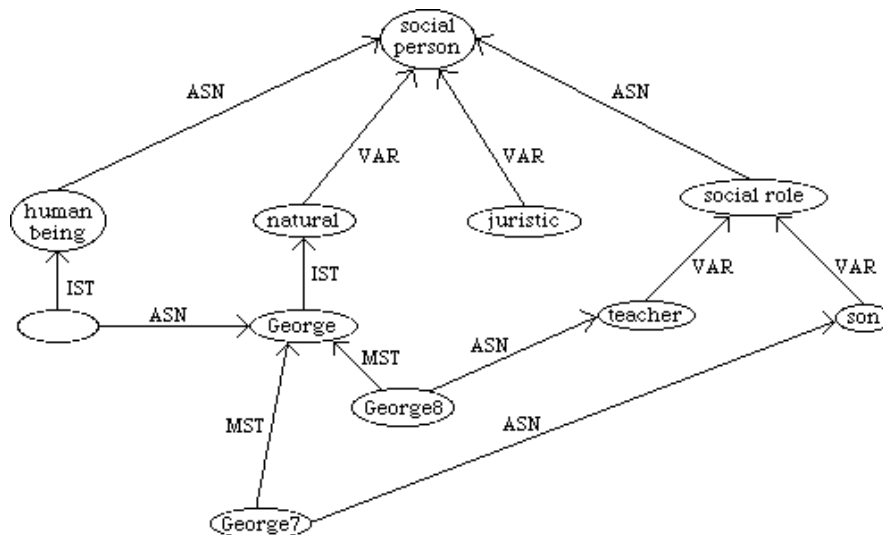


Figure 4: George as a Social Person

In Figure 4 George is treated as an instance (IST) of natural person, which is a particular variety (VAR) of social person; the other variety of social person, juristic person, refers to corporate entities, such as government agencies, and corporations (on natural and juristic persons see [10]). Note that George, as an instance of natural person, is distinctly different from George as an instance of human being. George the natural person is an assignment between a human being (node is unlabeled) and various social roles, in this case, teacher and son.

And we now have room to contemplate the irony of a submissive person playing a role which demands authority, or an assertive person playing a role which demands submission. The instance of person has personality traits attributed to him or her as an instance of person and those traits may be more or less consonant with the demands of any given social role. But, because roles and persons are in different paradigms, linked by assignment, we have separate control over inheritance in those paradigms. Roles inherit characteristics of roles only within the paradigm of roles, and the characteristics of roles are not inherited by the persons who are assigned to the roles.

We can provide a similar analysis of the concept of food. While there is no particular difficulty in considering food to be a variety of object, things become confusing when we consider varieties of food, such as apples and asparagus, venison and beef. For in addition to their links into the food paradigm, apples and asparagus need to be treated as subordinates of plant while venison and beef are subordinate to animal. By judicious use of assignment we can treat food as an assignment between some variety of plant or animal and a particular slot in an eating event. In Figure 5 apple is a variety of plant (actually, it is a component of an apple tree, which is a variety of plant).

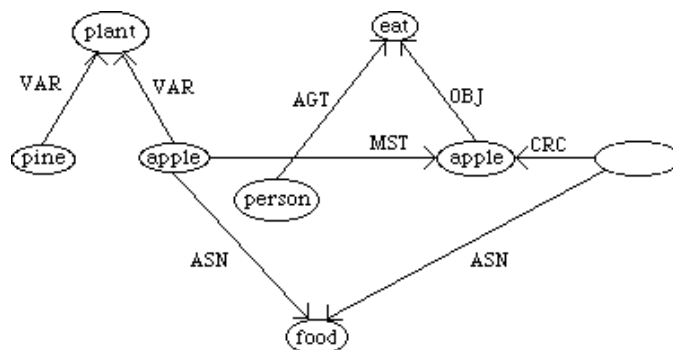


Figure 5: Food

A manifestation (MST) of the apple is being eaten. (A manifestation is a "time slice" of an object, that object during a particular stretch of time.) Attached to this manifestation of apple is a modifier, a circumstance (CRC), which is, approximately, edible - those attributes which allow an object to function in this role in this event. We have assignment links between the varietal apple and a node in the food paradigm and between the edibility circumstance and that same node in the food paradigm.

Note that the items in the food paradigm are likely to come from widely dispersed positions in plant and animal paradigms. Anthropological work on folk taxonomies [4, 8] suggests that folk classification of plants and animals is based primarily on the physical characteristics of the plants and animals, with visual appearance being the most important facet of those characteristics. In such a system foods will be dispersed throughout plant and animal paradigms for, even in a fairly simple diet, the elements of physical appearance which all foods could usefully inherit from common superordinate node are quite limited in comparison to what robin, pheasant, and hawk can inherit from bird or oak, maple, and even pine, can inherit from tree. By using assignment to link these widely dispersed items to a paradigm of foods we

make it possible to state generalizations about food and to create relatively compact high level structures, such as plans and scripts [30, 31] about food in general, with appropriate specification of detail for specific kinds of food.

But, what are the kinds of food? The plants and animals which are turned into food have positions in their respective paradigms, but are different paradigms. The difference between fruit and vegetable is one possibility. If we think of the difference between oranges, grapes, apples, and pears, on the one hand, and lettuce, cabbage, and spinach, on the other, we might gloss that distinction as one having to do with the part of the plant which is turned into food. But, corn, peas, and beans are vegetables, like cabbage, lettuce, and spinach, and, like oranges, grapes, apples, and pears (which are fruits), they are also the seed bearing portions of the plant, as opposed to being the leaves. So that particular account doesn't look promising. A more likely distinction would be based on their use as foods. Fruits are eaten as dessert, and only rarely with the main course. Vegetables regularly accompany the main course, and only rarely function as dessert. If food is itself a category defined by function, then its subcategories should differentiate along functional lines, which seems to be the case with fruit and vegetable. Another functional distinction was suggested by Schank [29], when, in objecting to inheritance structures, he pointed out that what is food for us may not be food for an animal, or vice versa.

While most of most of us know relatively little about animal diets, except for our pets, and hence don't devote much cognitive effort to classifying them, we do know (i.e. believe) that animals eat different things than we do, and their diets differ among themselves as well. Even where animals and humans share the same food sources (e.g. tuna fish for pet cats) the preparation is different, different enough to justify different nodes in a paradigm of foods. Thus a paradigm of foods is likely to differentiate between human and animal foods, and, among human foods, it will recognize distinctions such as breakfast vs. supper, main course vs. dessert, while among animal foods it may just differentiate by type of animal.

If we call the food paradigm a functional paradigm, then we can call the plant paradigm substantive, along with object, animal, human being, rational soul, and so forth (Figure 1). The nodes in functional paradigms are all linked by assignment to particular slots in events while the nodes in substantive paradigms have no such links. Other examples of functional paradigms would be toys and tools, which are objects used as toys or tools because of they have appropriate functional properties. In general, all objects, as opposed to actions, events, or properties, are organized into substantive and functional paradigms linked by assignment.

Having established, in broad outline, the nature of ontological structure, we can now consider ontological transitions, which are events that operate on the ontological structure of objects.

Paradigmatic and Ontological Transition

Some operations involving ontological structure ultimately conserve but change paradigmatic structure. Let us call these operations paradigmatic transitions. Operations which change ontological structure will be called ontological transitions. Ontological transitions are either compositions, in which an object's ontological structure is augmented, or decompositions, in which ontological structure is diminished.

As an example of a paradigmatic transition, consider the caterpillar and the butterfly (Figure 6). The large rectangles are supernodes [20] called modalities [19].

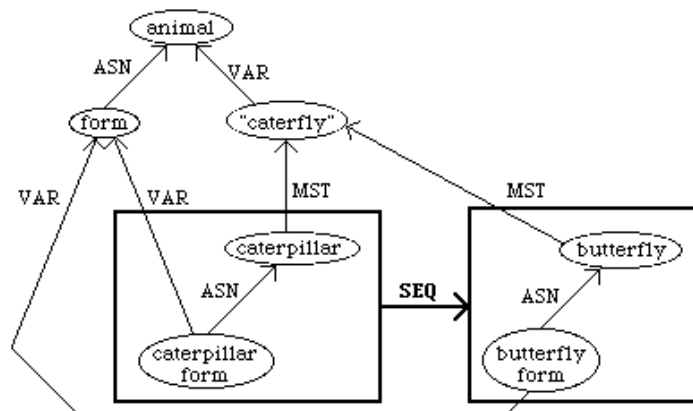


Figure 6: Caterfly

Modalities localize objects and events in time and space; they are the building blocks of episodic memory [32]. The objects and events so localized are the contents of the modality. Here we have two manifestations (MST) of "caterfly," one of which is a caterpillar and the other a butterfly. (Recall that manifestations are time slices of objects.) The second manifestation comes after the first in temporal sequence (SEQ). These two manifestations have the same assignment structure, which they have inherited from their common paradigmatic superordinate, "caterfly," which has, in turn, inherited it from the root of the animal paradigm (note that the assignment structure has been abbreviated for simplicity). The manifestations differ, however, in the forms which are assigned to their bodies, with the caterpillar having a "caterpillar form" and a butterfly having a "butterfly form." Its not an account of metamorphosis which would give a biologist much satisfaction, but it is probably adequate for a ten-year-old, whose ontology in this arena is much less sophisticated than the biologist's. And it is an account which conserves the essential continuity between the butterfly and the caterpillar, the identity, through the change in form.

A similar account could be given of the acorn and the oak, or of the ugly duckling and the swan. Hans Christian Anderson's story of the ugly duckling and the swan, however, is not, in our culture, simply one of shape change. It is a story about growing up, about the difference between child and adult. More precisely, the story of the ugly duckling which grows up to be a graceful swan is offered to children as an analogy in which they are to see their present life as that of the ugly duckling's and are asked to image a future in which they will be graceful swans. Anthropologists [14] have taught us that the transition from childhood to adulthood involves a change in social role. It is, in our terms, an ontological transition in which a social person exchanges one role for another. Human life is replete with such transitions, from single to married, from page to knight, from alien to citizen. In each case, one role is shed and another assumed.

What of exile? In that case, I submit, we have an ontological transition and not a paradigmatic transition. The point of exile, as a form of punishment, is to deprive the social person of all social roles, to strip them down to mere humanity, in King Lear's terms, "unaccommodated man . . . a poor, bare, forked animal." Exile is a decomposition. And christening is a ceremony in which a mere human infant is given an name, as an index of a social identity, and thereby brought into the social world. As such it is a compositional transition. Similarly, we can think of birth, or perahaps "coming to life" is more precisely what I have in mind, as when the soul joins the body and death as when it leaves - composition and decomposition. More humbly, animal or plant substance must be appropriately prepared if it is to become suitable food. And, while this, in part, reflects nutritional and digestive

requirements, there is also a ritualized cognitive component to food preparation [25, 26]. Animals and plants must be killed, decomposition, and then appropriately prepared, composition (with the edibility circumstance in Figure 5), as their flesh and fiber takes its place in the food paradigm.

This brings me to one final point about ontological structure. And this point is, in turn, about story grammars. In a recent review and critique of story grammars, Robert Wilensky [34] argues that most of the work on story grammars misses the essential point. Story grammars tell how something is packaged as a story, but not what sorts of thing a suitable for such packaging. To this end he sketches a theory of story points. Points are simply topics which make good stories, with "human dramatic situations" being a major class of topics. While it is not obvious that everything which is a human dramatic situation is either a paradigmatic or an ontological transition, I am sure that every paradigmatic or ontological transition in which humans participate is a dramatic situation. And, there are dramatic situations which, while they are not a paradigmatic or an ontological transitions, gain their drama from the possibility or threat of such transition - a nearly fatal boating accident, the groom abandoning the bride at the altar. Finally, as a sometime student of literature, I think it is safe to say that all of the stories which find their way into literature classes are stories of paradigmatic or ontological transition (for an extensive, though informal, textual analysis along these lines see [1]). Whether or not all stories are stories of paradigmatic and/or ontological transition, all of the important ones are. Thus the concept of ontological transition provides a compact characterization of the most important classes of story points.

CIMWorld

CIMWorld is a representation of the world of Computer Integrated Manufacturing (CIM). At present CIM is but a gleam in the eyes of those who want to integrate robotics, Computer Aided Design, Computer Aided Testing, Materials Resource Planning, and so forth, into one large, complex, and coherent automated factory. The CIM Program at RPI (Rensselaer Polytechnic Institute) is building a prototype CIM installation in which the machining and assembly of small machine parts and assemblies will be completely automated and is developing a general methodology for CIM. One problem which is obvious is simply that of keeping on top of the large and heterogeneous pile of information needed for designing and implementing CIM systems. An intelligent workstation would help with this problem. To that end, we have explored a knowledge representation language in which we can conveniently represent CIM concepts. This section explores some of the ontological aspects of CIMWorld, in particular, the ontology of parts and products.

Figure 7 is a fragment of the ontological structure of CIMWorld; it has much the same structure as Figure 1.

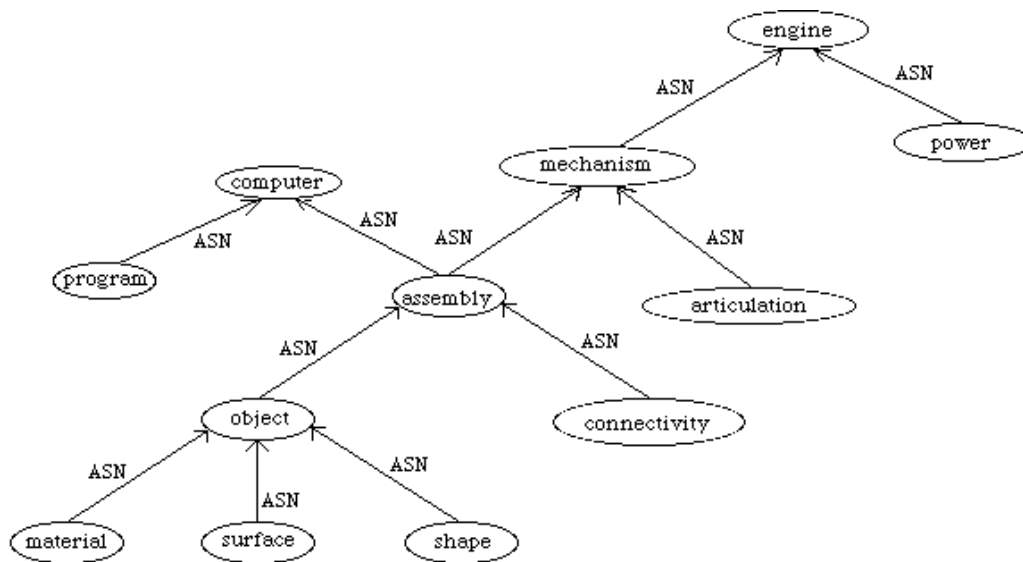


Figure 7: CIM Ontology

The object node in this diagram is a simple physical object having substance, surface, and shape as aspects of its ontological structure. Shape is the most complex of these aspects, requiring complex geometrical description coupled to a propositional frame [28]. The shape will have components, such as edges and vertexes or even component shapes. The substance of the part is the stuff of which the object is made; it has properties of mass, density, ductility, heat conductivity, etc. Finally, the surface must be considered separately from substance and shape because different types of process apply to it. The surface may be painted or plated, and/or ground to set specifications. This doesn't affect the shape or the nature of the substance.

An assembly has a different ontological structure than a primitive part, which is to say that an assembly is not merely a complex part. There is more to it. An assembly is an assignment between a part and a connectivity structure. To think of an assembly as a part, first imagine shrinking an envelope around the assembly. The resulting shape/surface/substance triple is the assembly as part. Its shape and surface might be quite complex (think of a automobile engine as an assembly which is part of the automobile), and its substance heterogeneous (e.g. rubber, plastic, three kinds of metal, etc.). This part is, of course, a complex part. As such it has components, the simpler (perhaps even primitive) parts which make it up (Figure 8). Simple objects do not have components, though their shapes do. Assemblies have objects and/or assemblies (considered as objects) as components.

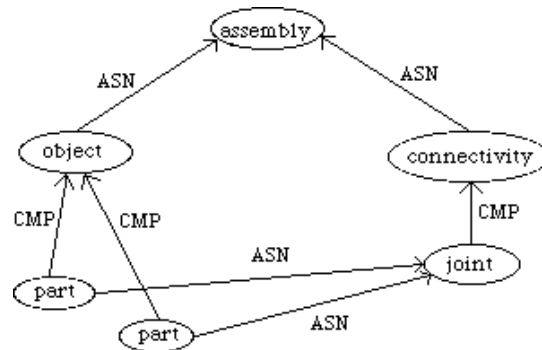


Figure 8: Assignment Structure of an Assembly

The connectivity structure is much subtler. Some of us are fond of saying that "the whole is greater than the sum of the parts." That statement is true of many things, including many mechanical assemblies; but it is also vague. What is in there besides the parts? For assemblies it is the connectivity structure. Connectivity structure arises because assemblies are made of parts. But it cannot be identified with the componentiation structure which constitutes part/whole relationships. Connectivity structure can be inferred from a complete representation of the spatial relationships between components, but it is not explicitly present in that representation. One way to think of connectivity structure is to imagine a graph in which joints between parts are represented by nodes (Figure 8). The parts incident on the joint are linked to it by assignment arcs. Assemblies in which e.g. all the parts but one are attached to that one common part (as in several gears attached to a single shaft) will have a different connectivity structure from assemblies in which e.g. each part participates in only two or three joints (e.g. a cylinder and connecting rod). In the process of assembly as parts are connected together the joint nodes onto which they are connected are satisfied. The process continues until the connectivity graph is completely satisfied, at which point the assembly is complete.

The assignment structure of an assembly is not, however, quite adequate for describing the structure of a mechanism in which parts move relative to one another. The paths of linear and rotary motion are as intrinsic to the structure of a mechanism as the connectivity structure is to an assembly. Consequently we can think of a mechanism as an assignment between an assembly and an articulation structure (Figure 7) in which nodes represent axes of rotary motion and paths of linear motion, with the appropriate junctions in the connectivity structure (of the assembly) being linked to these nodes by assignment. These articulation structure nodes are likely to be defined in the external base in different ways depending on whether we are dealing with the design database (where motion paths need to be present in some way so we can simulate the motion of the parts) or with a program which controls the mechanism (as in the program for a numerically controlled machine or for a robot).

Now consider engine, which is an assignment between a mechanism and a source of power (e.g. steam, fossile fuel, electricity). Mere mechanisms must be externally powered, water turns the wheel, a hand turns the crank. Engines have their own power; they are thus autonomous in a way that mechanisms are not. It would be a mistake, however, to think that the assignment link between power and engine somehow constitutes a representation of an explanation of how engines use power. That link no more explains the operation of engines than the assignment link between vegetative soul and plant explains what makes plants alive. And, just as people talk of plants and life force without knowing an modern biology, so people talk of engines and power without knowing any classical or modern physics. The point of asserting that engines are an assignment between mechanisms and power is simply to note that engines are not mere mechanisms. They are different, and that difference is attributable to their sources of power.

We should not, however, think that the domain of engines contains only those things we call motors. It also includes motor driven machine tools - such as lathes and drill presses. These machines include motors as components which provide the power to move appropriate mechanisms in appropriate ways; but we do not, merely by combining engines and mechanisms in ingenious ways, thereby generate new ontological domains. Thus complex machine tools can be considered to be varieties of engine. To use "engine" in this way is to make it a technical term whose meaning, in this context, doesn't quite match the meaning it has in ordinary language.

The ontological assertion that a computer is an assignment between an assembly (of chips) and an program is comparable to the assertion that an engine consists of a mechanism and a source of power. Even if we forget that real computers have mechanisms and engines in their disk drives, there is surely much more to it than that (see, for example, the articles by de Kleer [12] and Williams [35] on qualitative reasoning about electronic circuits). The essential

instance being the same in both cases.

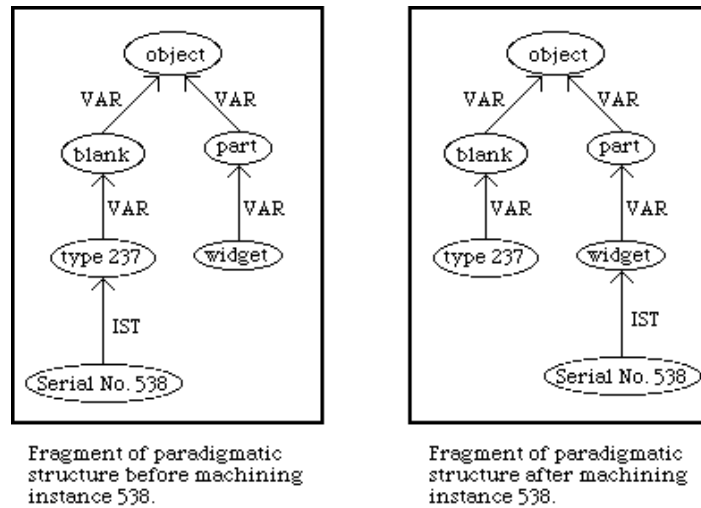


Figure 10: Instance Migration

A similar analysis applies to the process of putting an assembly together. At the beginning we have a bunch of unconnected parts and/or (sub)assemblies. The, two at a time, parts and subassemblies are jointed until finally the assembly is complete. This is a bit more complicated because it requires that we represent the difference between a satisfied and an unsatisfied slot the connectivity structure for an object. Figure 11 indicates the essentials. The supernodes on the left are modalities, and the nodes within them are therefore all manifestations. The supernodes on the right are, following David Hays [5] gnomons. Gnomonic structure keeps track of the reality status of objects, which encompasses not only traditional logic and quantification (compare Hendrix's [20] discussion of the orthogonal partitioning approach to quantification) but also such distinctions as that between real events and imaginary events. In Figure 11 the framis and the glamis are components of the blivit assembly. Above, the gnomonic predicate "real" applies only to the framis and the glamis. The connectivity structure and the assembly are imaginary. In the diagrams below, in which the modality is temporally after (SEQ, sequence) the upper modality, the connectivity structure, and therefore the assembly, is real. That is what the operation of assembly does; it converts an imaginary instance of connectivity into a real instance of connectivity.

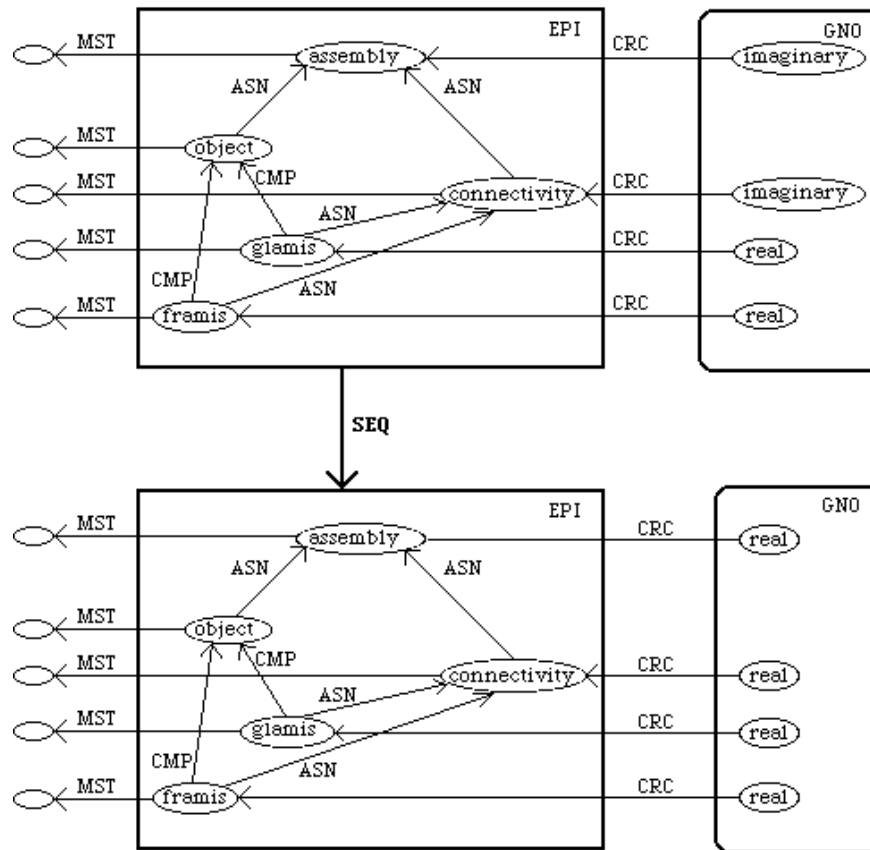


Figure 11: Satisfaction of Connectivity

From an informatic point of view we thus have the following characterisations:

- Machining has the effect of moving an instance from one type to another.
- Assembly has the effect of converting an imaginary connectivity structure into a real one.

These statements characterise, not the physical processes of machining and assembly, but the structure of the representations through which informatic processes regulate and control machining and assembly in a CIM environment.

Now, examine Figure 12, which shows that CIMWorld consists of InfoWorld (Informatic World) and PhysWorld (Physical World). CIMWorld is the entire representation domain, while InfoWorld and PhysWorld are two major partitions, called Realms of Being, within CIMWorld. When the representation is extended to include the financial and marketing functions of CIM, other partitions will probably be needed to accommodate them. Realms of Being are supernodes which include some connected portion of the assignment tree, along with the attributes and actions which are appropriate to the entities in that portion of the three.

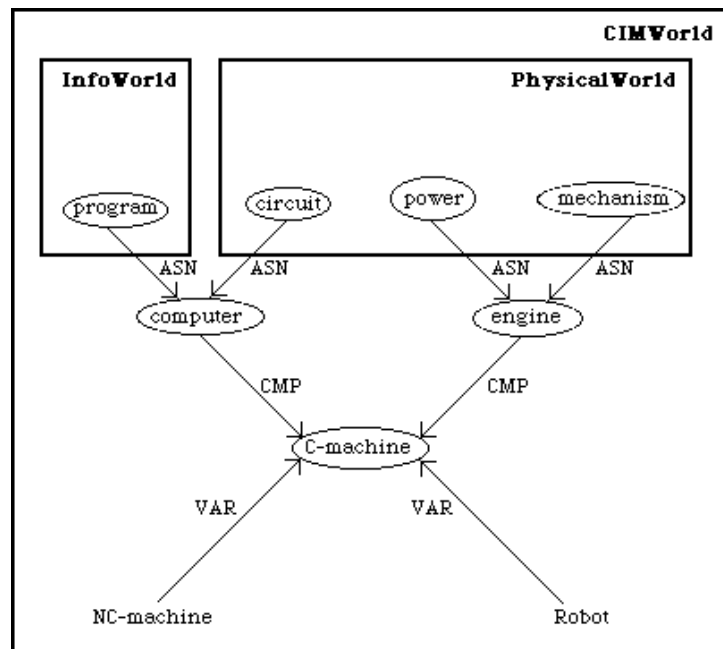


Figure 12: CIMWorld, PhysWorld, InfoWorld

Thus the PhysWorld Realm is a supernode which includes every physical domain in the ontology and every action which is specified only over physical objects - cut and join would be in PhysWorld. Control, and any of its varieties, however, would not be in PhysWorld. While physical entities are controlled, informatic entities are doing the controlling. The participation of informatic entities in control actions forces those actions into another realm, InfoWorld. If needed, PhysWorld could be differentiated into Realms specific to objects, to assemblies, to mechanisms, and to engines. InfoWorld, then, would consist of informatic entities, attributes, and actions. Notice that some entities don't really exist in either PhysWorld or InfoWorld. Computers straddle the two Realms. C-machines (computer machines) are any machine which includes a computer as a component; robots and numerically controlled machine tools are the two major classes of such machines. C-machines also straddle the two Realms.

Within InfoWorld we have to represent the world of dataprocessing to whatever degree of detail is useful. More specifically, we need to represent it in enough detail so that CIMWorld can be used in designing CIM databases, so that CIMWorld can function as a data dictionary [11] for CIM. Thus we need an informatic ontology which includes representations of data types, languages, algorithms, and programs, etc.

Now we can state one requirement of CIM design, that of completeness:

- Everything that is in PhysWorld must be represented in some database in InfoWorld.

There is nothing particularly subtle about this requirement but, when you consider the diversity and complexity of information involved in manufacturing, the completeness requirement is a bit daunting. Every drill bit, every machine tool, every raw workpart, every manufacturing operation, and so forth, must be represented.

But this is not enough. The relationships between InfoWorld and PhysWorld must be causally coherent. It is not clear just what causal coherence is, and it certainly is not a function of the CIMWorld representation. However, we are hoping that the CIMWorld representation will give us enough control over CIM design information so that it can function as a cognitive helicopter, lifting us high enough over the trees of design detail so that we can see the shape of the forest.

It is clear, for example, that CIM requires various sorts of feedback loops. Once a part has been machined it must be inspected to determine whether or not it meets design specifications. If it doesn't meet specifications then it must be rejected and the reason for the failure must be determined. It might be a simple matter of replacing a worn cutting tool. The problem, might, however, be more serious, requiring a major alteration in the process plan for manufacturing the part or even in the design of the part itself - perhaps the original design simply cannot be readily manufactured. Now, it is not up to CIMWorld to make any of these determinations - that will be done by applications programs and by human operators within CIM. But the CIM design, which is the provenance of CIMWorld, must make the relationships between these tasks and processes explicit. They must exhibit a causally coherent structure. If, in the design of a CIM system, we have a manufacturing task whose results are not verified by inspection, then causal coherence has been violated. If we have provision for inspection, but no provision for using inspection results to make appropriate modifications, then causal coherence has been violated.

The general idea is to formulate a concept of causal coherence within CIM which operates throughout the CIM environment, from shop floor systems operating on time scales of seconds, minutes, and hours to financial and marketing systems operating on time scales of weeks, months, and years. As I remarked above, formulating this concept is not a goal of the CIMWorld project. But, formulating such a principle requires a coherent view of the CIM environment; and I think that a knowledge-based design tool is essential to getting such a view. Since causal coherence, whatever it is, involves the interaction between various Realms of Being within CIM, an explicit representation of ontology is essential in this tool.

Multiple Ontologies

Within AI and cognitive science questions of ontology have most consistently cropped up in work on representing knowledge of physical systems [6, 16, 17]. Some of this work concerns the knowledge experts have in their specialty while other work is concerned with naive user's knowledge of technical domains. It is this latter work which I want briefly to consider.

The general finding is that novices and experts have different ontologies. It is not that the novice has a poor grasp of the ontology appropriate to some given domain; but that novice and expert structure a domain in fundamentally different ways. Thus Larkin, in her study of representation of physics problems, asserts that novices have only a *naive* problem representation while experts have, in addition, a "*physical* representation that contains fictitious, imagined entities such as forces and momenta" [24: 75]. Clement [9] and McClosky [27] find that naive theories of motion are much like pre-Newtonian mechanics. Wiser and Cary [37: 267], looking at a similar phenomenon in thermodynamics, explicitly suggest that "the naive-expert shift involves the same kinds of conceptual change as does theory change in the history of science."

We are thus back where we began, with salt and sodium chloride. The concept of salt is defined in a naive ontology while that of sodium chloride exists in a sophisticated ontology. The physical chemists among us understand what sodium chloride is. To the rest of us, it is just salt - and, I would imagine, that it is just salt to the physical chemist when he or she ceases to operate within the role of professional chemist and becomes, instead, merely a hungry person seasoning a packet of fast-food french fries. My little homily about salt and sodium chloride can be multiplied many times over. Logicians have long been fond of a homily about the Morning Star and the Evening Star, which are not stars at all, but the planet Venus. The point of their homily isn't about ontology; but the mere existence of the homily belies an ontological issue. For the astronomy in which those two objects are the same is working from an ontology of celestial objects and motions which is different from that used in an astronomy where those objects are distinct. And then we have the difference between *homo sapiens sapiens* and the

human race. The former is a term in a specialized biological ontology which sees no ontological distinction between the species *homo sapiens sapiens* and any other animal species. But "human race" is a term in a naive ontology which does see an ontological distinction between human beings and animals. And, to give a final example, I suspect that a skilled auto mechanic is operating with an ontology for mechanical objects - assemblies, mechanisms, and engines - which is different from that of someone with no particular skill in that domain. The difference might not be so dramatic as the difference between a layman and a physicist, but it is just as real.

Where do these different ontologies come from? The answer to that is simple: They come from the human mind. But there is another question whose answer is not so simple: What leads the human mind to create a new ontology? Let us approach this question indirectly.

From Harry Jerison [21] I adopt the view that an animal's nervous system gives it an ontology [3]. I am just enough of a biologist to be content with the view that human beings are animals and that, therefore, we have a nervous system that gives us an ontology. The structure of that basic ontology is not at all obvious, but Frank Keil's [22] work seems to be about that basic ontology. The puzzle is: Where do all of the specialized ontologies we've created in the last 5,000 years come from? It would seem that the ontology given in the human nervous system is open-ended, that we have the capacity to fashion new ontologies. Any explanation of how we do that is surely going to require a robust theory of abstract conceptualization [2, 18, 19]. That theory seems to me to be a bit beyond us, but I would like to conclude by suggesting a clue. The clue I offer is not offered as a proposal, no matter how tentative, but as an example of the sort of proposal which follows from an explicit account of ontological structure. This proposal follows from my particular account of ontological structure. If that account is no good, then proposal falls as well. But, a similar proposal should be possible within the more adequate account of ontological structure which replaces mine.

As background, let us consider the ubiquity and importance of conservation principles in science [13], which suggests that science abhors ontological transitions. Conservation principles tell us that, like the case of the butterfly and the caterpillar, beneath superficial change there is deeper identity. It is not that butterflies and caterpillars are the same; they are not. But one can become another. And the difference is only paradigmatic, not ontological. Similarly, it is not that electricity and heat are the same. But one can become the other, as when an electric current is passed through a resistor; while energy changes from one kind to another, the total energy in the system remains the same - that is conservation. Conservation principles define the boundaries of a particular ontological domain, asserting that things can change form within that domain; which means that, informatically, we have a paradigmatic transition. But things cannot leave or enter the domain.

Now, we are obviously a long way from the concreteness of the butterfly and the caterpillar. And, since I have provided no explicit mechanism for constructing more abstract ontologies, such as the one in which the conservation of energy can be expressed, it is hard to see just what this suggestion entails. Nonetheless, I find it suggestive that one of the earliest fragments of Greek philosophical thought seems to be a conservation principle. The fragment is attributed to Anaximander, who was born around 611 or 619 B.C. and it goes like this [33]:

The Unlimited is the first-principle of things that are. It is that from which the coming-to-be [of things and qualities] takes place, and it is that into which they return when they perish, by moral necessity, giving satisfaction to one another and making reparation for their injustice, according to the order of time.

This is, to say the least, rather obscure. But it is clear that everything is to be considered as being grounded in the Unlimited. Change - coming-to-be - is seen as injustice; but injustice is not permanent. And change, for the Presocratics, certainly included the passing of the seasons, night and day. Anaximander seems to be saying that the existence of something in some

particular form, at a given time, requires that, at another time, it exist in another form. For that is what justice demands. What is, in our terms, conserved, is the Unlimited. It is the nature of the Unlimited which requires that an appearance in one form be balanced by appearances in other forms. Thus everything, no matter what form, is an activity of the Unlimited.

If science, then, works within domains where we have only paradigmatic transitions, perhaps the anomalies that have figured so prominently in recent discussions of scientific revolution [23] betray the inadequacy of an ontology. Anomalies are things that occur but which cannot be accommodated within the current scientific theory. What constitutes an anomaly? The ontological structure of a domain is a structure of assignments that constitute the substantive and, through linkage to actions, the functional paradigms of objects in the domain. An anomaly occurs when objects undergo actions that do not conserve ontological structure, that are not paradigmatic transitions. Anomalies are that which have to be represented as ontological transitions, which science abhors.

The occurrence of anomalies indicates the existence of a new domain. That domain now has to be explored. The role of analogy in scientific thinking has been much discussed (for example, in [15, 17]). Here is a place to use analogy. Take some source domain that you understand and structure a new domain by analogy to the source. That is, you create objects in the new domain and identify them with objects in the source domain. Then you operate on those new objects in ways which are analogous to the operations in the source domain. When you arrive at a set of objects and operations in the new domain which do conserve ontological structure, your job is done. The objects and operations which satisfy this criterion might be far from the ones you initially proposed, by analogy. But analogy got you started.

This is surely too vague to count as an explanation. The important point is simply that an explanation is going to need an explicit account of ontological structure. I have sketched a fragment of such an account. That fragment relates entities to one another through assignment trees and links entities to actions to create functional paradigms. Within that account we can represent the critical distinction between paradigmatic and ontological transitions. For much of human thought clusters around ontological transitions. Myths explain the origins of everything, the sun and the moon, the earth, human beings, gnats, table manners, and roast pig. And science attempts to eliminate ontological transition by constructing new ontologies in which ontological transition is eliminated in favor of paradigmatic transition. Regardless of the details of representation or implementation, ontological structure is thus an important aspect of knowledge representation.

ACKNOWLEDGEMENTS

This work was partially supported by the Computer Integrated Manufacturing Project of the Center for Manufacturing Productivity and Technology Transfer at Rensselaer Polytechnic Institute. I would like to thank Dr. Geoffrey Goldbogen for his encouragement and support and David Ferrucci for his valiant efforts at implementing my all too obscure ideas. Many of the ideas in this paper evolved in conversations with these men and, above all, with Dr. David G. Hays, friend, colleague, and teacher for over a decade.

References

1. Benzon, William, *Sir Gawain and the Green Knight* and the semiotics of ontology, *Semiotica* 2 (1977), 267 - 293.
2. Benzon, William, Cognitive science and literary theory, SUNY at Buffalo Ph.D. Thesis, Buffalo New York, 1978.
3. Benzon, William, and Hays, David G., Principles and development of natural intelligence, manuscript.
4. Berlin, Brent, Breedlove, Dennis E. and Raven, Peter H., General principles of classification and nomenclature in folk biology, *American Anthropologist* 75 (1973) 214 - 242.
5. Bloom, David, and Hays, David G., Designation in English, in J. Hinds (Ed.) *Anaphora in Discourse* (Linguistic Research, Inc., Edmonton, Canada, 1978), 1 - 68.
6. Bobrow, Daniel G., Qualitative reasoning about physical systems, *Artificial Intelligence* 24 (1984) 1 - 5.
7. Brachman, Ronald J. and Schmolze, James G., An overview of the KL-ONE knowledge representation system, *Cognitive Science* 9 (1985) 171 - 216.
8. Brown, Cecil H., Folk botanical life-forms: Their universality and growth, *American Anthropologist* 79 (1977) 317 - 342.
9. Clement, John, A conceptual model discussed by Galileo and used intuitively by physics students, in: D. Gentner and A. L. Stevens (Eds.), *Mental Models* (Lawrence Erlbaum, Hillsdale, 1983) 325 - 340.
10. Coleman, James S., *Power and the Structure of Society* (W.W. Norton, New York, 1974).
11. Date, C.L., *An Introduction to Database Systems*, Second Edition (Addison-Wesley, Reading, Mass., 1977).
12. De Kleer, Johan, How circuits work, *Artificial Intelligence* 24 (1984) 205 - 280.
13. Feynman, Richard, *The Character of Physical Law* (MIT Press, Cambridge, Mass. 1965)
14. Gennep, Arnold van, *The Rites of Passage*, Translated by M. B. Vizedom and G. L. Caffee (University of Chicago Press, Chicago, 1960 [1909]).
15. Gentner, Dedre, and Gentner, Donald R., Flowing waters or teeming crowds: Mental models of electricity, in: D. Gentner and A. L. Stevens (Eds.), *Mental Models* (Lawrence Erlbaum, Hillsdale, 1983) 99 - 130.
16. Gentner, Dedre, and Stevens, Albert L. (Eds.), *Mental Models* (Lawrence Erlbaum, Hillsdale, 1983).
17. Greeno, James G., Conceptual entities, in: D. Gentner and A. L. Stevens (Eds.), *Mental Models* (Lawrence Erlbaum, Hillsdale, 1983) 227 - 252.
18. Hays, David G., The meaning of a term is a function of the theory in which it occurs, *SIGLASH Newsletter* 6 (4) (1973) 8 - 11.
19. Hays, David G., *Cognitive Structures* (Human Relations Area Files Press, New Haven, 1981).
20. Hendrix, Gary G., The representation of semantic knowledge, in Donald E. Walker (Ed.), *Understanding Spoken Language* (North-Holland, New York, 1978).
21. Jerison, H.J., Paleoneurology and the evolution of mind, *Scientific American* 234 (1976) 90 - 101.
22. Keil, Frank C., *Semantic and Conceptual Development* (Harvard University Press, Cambridge, MA., 1979).
23. Kuhn, Thomas S., *The Structure of Scientific Revolutions* (University of Chicago Press, Chicago, 1962).
24. Larkin, Jill H., The role of problem representation in physics, in: D. Gentner and A. L. Stevens (Eds.), *Mental Models* (Lawrence Erlbaum, Hillsdale, 1983) 75 - 100.

25. Leach, Edmund, Anthropological aspects of language: Animal categories and verbal abuse, in: W.A. Lessa and E.Z. Vogt (Eds.), *Reader in Comparative Religion* (Harper and Row, New York, 1965).
26. Lévi-Strauss, Claude, *The Raw and the Cooked* (Harper and Row, New York, 1969).
27. McClosky, Michael, Naive theories of motion, in: D. Gentner and A. L. Stevens (Eds.), *Mental Models* (Lawrence Erlbaum, Hillsdale, 1983) 299 - 324.
28. Minsky, Marvin, A framework for representing knowledge, in: P. Winston (Ed.), *The Psychology of Computer Vision* (McGraw-Hill, New York, 1973) 211 - 277.
29. Schank, Roger C., The structure of episodes in memory, in: *Representation and Understanding* (Academic Press, New York 1975) 237 - 272.
30. Shank, Roger C., *Dynamic Memory* (Cambridge University Press, Cambridge 1982).
31. Schank, Roger C. and Abelson, Robert P., *Scripts, Plans, Goals, and Understanding* (Lawrence Erlbaum, Hillsdale, N.J., 1977).
32. Tulving, Endel, Episodic and semantic memory, in: Endel Tulving and Wayne Donaldson (Eds.), *Organization of Memory* (Academic Press, New York, 1972) 382 - 403.
33. Wheelwright, Philip (Ed.), *The Presocratics* (Odyssey Press, New York, 1969) 54.
34. Wilensky, Robert, Story grammars versus story points, *The Behavioral and Brain Sciences* 6 (1983) 579 - 623.
35. Williams, Brian C. Qualitative analysis of MOS circuits, *Artificial Intelligence* 24 (1984) 281 - 346.
36. Winograd, Terry, Frame representations and the declarative-procedural controversy, in: *Representation and Understanding* (Academic Press, New York 1975) 185 - 210.
37. Wiser, Marianne, and Carey, Susan, When heat and temperature were one, in: D. Gentner and A. L. Stevens (Eds.), *Mental Models* (Lawrence Erlbaum, Hillsdale, 1983) 267 - 298.